

REQUEST FOR PROPOSAL - DEMO PACKAGE

Steam Conditioning Valve Package

Steam Header Pressure Letdown and Desuperheating Upgrade

FICTIONAL CUSTOMER PROFILE

Meridian Combined Cycle Power and Utilities

INDUSTRY

Power / Utilities

ISSUE DATE

29 May 2026

BID DUE DATE

03 July 2026

COMMERCIAL CONTEXT

Estimated package value: INR 24.2 Cr

PACKAGE TYPE

Severe-service control valve proposal

Demo-safe fictional RFP package created for WinsProposal. Customer names are fictional industry-style accounts and do not represent official documents, endorsements, trademarks, or partnerships.

1. Purchaser Profile and Project Background

Meridian Combined Cycle Power and Utilities is a fictional power-sector owner profile used to represent a rigorous utility customer with high expectations for thermal performance, noise attenuation, testing, and documentation.

The purchaser requires steam conditioning valves for pressure letdown and temperature control across startup, bypass, and continuous operation. The package must address desuperheating, spray water conditions, noise attenuation, thermal cycling, actuator/control accessory requirements, and inspection/testing deliverables.

Proposal expectation: Supplier response shall show engineering proposal intelligence: extracted requirements, compliance/TBE readiness, severe-service risk recognition, knowledge reuse, deviations, review owners, and management-ready documentation.

2. Scope of Supply

- Supply steam conditioning valves and spray water control accessories.
- Provide proposal-stage steam sizing, thermal balance assumptions, and noise/thermal cycling risk narrative.
- Provide actuator and control accessory selection basis.
- Provide compliance matrix, TBE responses, QA/QC plan, and manufacturing data record.
- Provide list of engineering clarifications needed for final validated design.

3. Line Items

Tag / Ref.	Item	Qty	Indicative Size / Class	Service Notes
PCV-SC-4101A/B	Steam conditioning valve	2	NPS 8 x 12, ASME 1500/600	Pressure letdown and desuperheating
TCV-SW-4101A/B	Spray water control valve	2	NPS 2, ASME 900	Spray water regulation
ACC-SC-4101	Actuator and accessory package	1 lot	Pneumatic control	Positioner, solenoid, airset, switches
DOC-SC	Engineering and QA dossier	1 lot	MDR package	Datasheets, assumptions, ITP, certificates

4. Process Conditions

Input	Requirement / Value
Fluid	Superheated steam
Inlet pressure	125 barg
Outlet pressure	42 barg
Inlet temperature	520 C
Outlet temperature target	290 C
Spray water pressure	70 barg
Duty	Startup, bypass, and continuous conditioning cases

5. Technical Specification

- Supplier shall propose severe-service steam pressure letdown trim with staged pressure reduction and noise attenuation.
- Supplier shall provide desuperheating narrative including spray water pressure, quality, atomization assumptions, and outlet temperature margin.
- Supplier shall identify choked-flow, acoustic fatigue, thermal cycling, vibration, and downstream piping risk assumptions.
- Supplier shall provide actuator/accessory basis for control stability, fail action, instrument air, and functional testing.
- Supplier shall identify final process data required for certified sizing, thermal balance, and acoustic guarantee.
- Supplier shall separate proposal-stage indicative engineering from final validated design.

6. Compliance Requirements

- ISA 75.01 / IEC 60534 sizing awareness.
- ASME B16.34 pressure-temperature considerations.
- Project steam conditioning and noise specifications.
- Project inspection, hydrotest, seat leakage, and functional test requirements.
- Client manufacturing data record and final documentation procedure.

7. Inspection and Testing Requirements

- Hydrostatic shell test.
- Seat leakage test.
- Functional test of actuator/accessory package.
- Spray water component inspection.
- Material certificates and dimensional inspection.
- Client witness points for pressure test and final inspection.

8. Required Proposal-Stage Engineering Intelligence

Engineering Narrative Required

- Extract process conditions and classify application/service.
- Provide preliminary valve configuration narrative.
- Provide severe-service risk assessment.
- Provide cavitation/flashing/choked-flow narrative where relevant.
- Provide noise/vibration risk narrative.
- Provide trim, material, actuator, and accessory notes.

Calculation Summary Required

- State process inputs and assumptions.
- Classify pressure drop severity.
- Provide indicative Cv/Kv placeholder only where safe.
- List risk flags and missing data.
- List validation checklist and standards awareness.
- State that final sizing/design requires engineer validation.

9. Technical Bid Evaluation Template

Evaluation Tag	Purchaser Requirement
Material	High-temperature pressure boundary material per ASME and thermal cycling review.
Trim	Staged severe-service steam trim with low-noise design basis.
Spray	Desuperheating arrangement to be defined by heat balance and spray water data.
Actuator	Pneumatic actuator/accessories sized for fail action and stable control.
Testing	Hydrotest, seat leakage, functional testing, and witness points.
Documentation	Datasheets, GA, assumptions, ITP/QAP, certificates, and deviations.

10. Evaluation Criteria

Criterion	Weight
Steam conditioning technical response	30%
Noise and thermal risk handling	20%
Compliance and inspection readiness	20%
Delivery	10%
Commercial value	20%

11. Possible Deviations and Clarifications

- Noise guarantee requires final flow cases, outlet pipe geometry, and project noise criteria.
- Desuperheating performance requires confirmed spray water pressure, temperature, quality, and control philosophy.
- Final Cv/Kv and spray water demand require validated steam sizing and heat balance.

12. Engineering Review Points

- Confirm steam flow cases for startup, bypass, and continuous operation.
- Confirm spray water pressure, temperature, quality, and turndown.
- Confirm outlet temperature target and minimum superheat margin.
- Confirm project noise limits and outlet piping geometry.
- Confirm actuator fail action and control accessory requirements.

13. Commercial and Delivery Assumptions

- Supplier shall state drawing submission lead time, manufacturing lead time, test duration, and shipping readiness.
- Supplier shall identify long-lead materials, special trims, actuator accessories, and inspection hold points that may affect schedule.
- Supplier shall provide itemized pricing by line item and list optional compliance or testing costs separately.
- Supplier shall include a deviation register with technical, commercial, and schedule impact.
- Proposal validity shall be not less than 60 days from bid due date.

Mandatory wording: Any engineering analysis in the supplier proposal shall clearly state: "Preliminary proposal-stage engineering estimate. Final sizing/design must be validated by qualified engineers using company-approved tools and standards."